

Employee Stock Ownership Plans and Workplace Safety

Abstract:

This paper analyzes the relationship between workplace health and safety outcomes and the presence of an Employee Stock Ownership Plan (ESOP) using several panel regressions and a differences-in-differences approach to understand whether the adoption of a participatory structure causes a change in injury and illness rates. I combine firm-level data from the Department of Labor with establishment-level data from the Occupational Safety and Health Administration covering 2016 to 2019. Over this four-year observation period, I find that ESOPs are generally associated with lower injury and illness rates per 100 full time workers. Additionally, I document evidence that adopting an ESOP may cause workplace safety conditions to significantly improve over time.

Introduction

Employee-owned companies have been shown to surpass traditional ownership structures on several positive performance measures, but does this exposure to shareholder compensation introduce a perverse incentive to overlook safety precautions? Or do employee-owners consistently secure the preservation of their health and human capital through their participation

in managerial decision-making? Employee-owned firms have long evidenced improvements to labor productivity, growth, efficiency (Blinder 1990; Kumbhakar and Dunbar 1993; Mitchell, Lewin, and Lawler 1990; Park and Song 1995), employment stability and survival rates (Kruse 2002; Kurtulus and Kruse 2016; Core 2001; Kruse et al. 2010; Bryson et al. 2016), return on assets (O’Boyle, Patel, and Gonzalez-Mulé 2016), and employee wages and retirement benefits (Sesil et al. 2007).

While employee-owned firm structures enjoy a vast body of supportive literature regarding compensation and performance, studying how the incentives of employee ownership interact with workplace health and safety has received limited treatment. I seek to address this gap by analyzing workplace injuries and illnesses - a unique form of non-pecuniary compensation - in labor-managed companies. Diligent preservation of human capital is integral to a worker’s ability to earn income over time, and reducing workplace injuries and illnesses is an essential component of fairness in the labor market. Ensuring safe working conditions reduces turnover, absenteeism, and medical costs, contributing to employee-owned companies’ consistent successes.

The universe of employee-owned (hereafter, EO) structures is large, so the research contained here focuses on the most widely practiced and well-integrated form of employee ownership: the Employee Stock Ownership Plan (ESOP). An ESOP is a private pension plan which was first introduced in 1974 under the Employee Retirement Income Security Act (ERISA). Since then, ESOPs have ballooned in popularity. U.S. Department of Labor filings indicate there were 14.9 million total participants, and 10.8 million active employee participants, in ESOPs in 2022 (USDOL 2024: 62). The exact structure, details, and complementary participatory management practices vary widely from one company to the next (Logue, Greider,

and Yates 2018). Previous studies cover a range of topics including performance, risk propensity, stability, and employee participation.

My investigation is guided by the following question: How does the presence of an ESOP affect workplace health and safety outcomes? I construct a novel panel dataset using nationally representative administrative data over four years at both the establishment and firm level. I then use a suite of OLS estimation methods and an event study design to provide evidence on this relationship. This paper provides both a statistical association and causal relationship of meaningfully large reductions in the occurrence of injuries and illnesses with companies which have adopted an ESOP.

ESOP Research

From a theoretical perspective, the effect of an ESOP on health and safety outcomes is ambiguous; these outcomes may either be supported by participatory structures or worsened by higher performance expectations. There is strong reason to believe that these participatory structures will support health and safety outcomes. The first avenue is by improving employee retention. Survey results have revealed that the perceived influence from an ESOP is one of the strongest sources of job satisfaction (Buchko 1993), while population administrative data has evidenced increased employment stability at ESOP firms (Kurtulus and Kruse 2016). Both employment stability and job satisfaction directly lead to higher retention rates (Core 2001; Bryson et al. 2016; Kruse et al. 2010). Intuitively, this gives employees more time to effectively implement firm-specific health and safety practices and guidelines, potentially reducing incidence rates.

A second channel through which ESOPs may improve workplace safety is by reducing agency costs. Equity compensation aligns incentives between the employee and the firm. This alignment of incentives reduces the likelihood for employees to shirk work responsibilities - known as the free-rider or 1/N problem - and empirical studies have noted these performance improvements even with small amounts of employee ownership (Igalens and Roussel 1999; Oyer 2004; O'Boyle, Patel, and Gonzalez-Mulé 2016). These aligned incentives are also associated with an increased prevalence of employee co-monitoring (Kruse et al. 2010), which could lead to workers checking that their colleagues are following proper workplace safety measures, thus creating a less hazardous environment.

The third avenue is through participation in firm decision-making. ESOP employees are more likely than non-ESOP employees to participate in workplace decisions and board meetings (Logue, Greider, and Yates 2018; Bruner 1988; Rosen 2013; Kruse et al. 2010). Several empirical studies have found that EO firms have more egalitarian compensation distributions than their traditional counterparts (Abramitzky 2008; Burdín 2016; Dow 2018; Kremer 1997; Montero 2022; Alves et al. 2012), indicating that ESOP employees carry bargaining power to help them secure advantageous working conditions. Employees presumably value their health, safety, and wellbeing, and being involved in management decisions allows them to bargain for the protection of these values.

The reasons that health and safety concerns may be diminished by an ESOP are equally compelling. Results from compensating wage theory suggest that wages and workplace safety share a differential relationship. The above-market wages and equity-based compensation found in ESOPs may suggest that ESOP employees are expected – or prefer – to increase production schedules and take additional safety risks to achieve higher profits (Rooney 1992). Even when

their elevated compensation is viewed through the lens of efficiency wages, exposing workers to company profits can incentivize them to overwork despite workplace safety measures (Akerlof 1984).

The differences-in-differences approach has been often adopted among ESOP studies to mitigate endogeneity issues regarding the reasons why a firm might adopt an ESOP. Logue, Greider, and Yates (2018) attempted a total population survey of ESOPs in Ohio between June 1992 to May 1993, achieving an 83% response rate. They collected responses from 167 companies which confirmed their ESOP status. Major and minor reasons for adopting an ESOP included: philosophical commitment to employee ownership (87%), providing another employee benefit (83%), buying stock from a retiring owner (70%), participative management (64%), reduced borrowing costs (36%), expanding the business (29%), averting a takeover bid (22%), avoiding shutdown or job loss (13%). It should be noted that not all 167 survey respondents answered every question above. The fact that a philosophical commitment to employee ownership and interest in participative management are among the most cited reasons for adoption in so many firms highlight the concern for self-selection bias. These figures further emphasize the need to control for the variety of intentions behind ESOP adoption. For this reason, a differences-in-differences approach is also implemented here.

Workplace Safety Research

Worker health and safety has recently garnered increased attention in economic literature through the lens of financial investment limitations at cash-constrained companies. Cohn and Wardlaw (2016) find that financial frictions and cash constraints can limit investment into workplace safety. Zolotoy et al (2022) find that exposing rank-and-file employees to stock options improves workplace safety in cash-constrained firms using datasets similar to the ones

used here¹. Although both ESOPs and stock options are forms of equity compensation, only the former grants employees certain ownership-related privileges and rights at their workplace. This additional influence on organizational decision-making has potentially large consequences inside such a complex system.

Several interdisciplinary researchers and journals from Europe feature articles on workplace safety. Of particular relevance, van Kampen, Lammers, Steign, et. al. (2023) administered a survey on safety practices to 297 firms and found that 56% of respondents cited “Employee safety training focused on cooperation” as one of the top 3 most effective safety interventions, being adopted at 88% of the participating firms. The response rate for this intervention was the most cited effective intervention by a large margin. Pairing this with the heightened rates of employee co-monitoring found by Kruse et al. (2010), this finding bodes well for an ESOP’s safety environment.

The largest swath of research among all social sciences that directly concerns workplace safety can be found in the ergonomics literature. Ergonomists study the efficiency of humans in work environments; their work comprises a subfield of psychology. The ergonomist Pascale Carayon et al (2015) published a framework for describing workplace safety using a “sociotechnical systems approach”. Here, safety is described as an emergent phenomenon resulting from interactions between components in a complex, non-deterministic, dynamic system and from the environment in which that system exists.

This framework proposed a major departure from prior perspectives about workplace safety by placing the individual worker at the center of the system and looking at their behavior.

¹ We both use the PPP data, but Zolotoy et al use the ITA’s predecessor dataset, which covers a longer observation window and generally contains more complete data. The data used here are more recent.

It has since been widely adopted in applied ergonomics. This sociotechnical framework is too complex to be modeled empirically with this paper's data, but it overlaps conceptually in some important ways. First, it acknowledges the familiar tradeoff between productivity and safety, as suggested above regarding how ESOPs might spur workers to overlook safety measures. Second, it acknowledges that any effects resulting from changes made to these dynamic, chaotic systems will require time before safety emerges in a steady state. This second conceptual overlap leads me to explore whether an ESOP's effect exhibits hysteresis in the penultimate section of my methodology.

Given the theoretically ambiguous effect that an ESOP has on health and safety outcomes, an empirical setting will help to identify which effect dominates. In general, I find that ESOPs are statistically associated with fewer injuries and illnesses per 100 full time workers, and the adoption of an ESOP is followed by a reduction in incidence rates in line with the statistical association mentioned above. I outline the data and empirical methods which produced these results in the following sections.

Data

This study combines two datasets, each with four observation years, from 2016 to 2019. The first is the Private Pension Plan Research File (PPP) from the U.S. Department of Labor (USDOL). Second is the Injury Tracking App (ITA) from the Occupational Safety and Health Administration (OSHA). They are described sequentially below.

Private Pension Plan Research File (PPP):

Under the Employee Retirement Income Security Act (ERISA), private firms offering employer-sponsored pension plans must file Form 5500. United States Department of Labor

(USDOL) annually cleans the data, adds standardized variables, and then releases it publicly as the Private Pension Plan Research File (PPP). It provides population administrative data of firm level pension plans from 2000 to 2019, including data on the specific details of the plan structure, its financial state, and participation among employees. The benefit of using population data is the elimination of sample selection bias and survey-response bias. This dataset is used extensively not only in ESOP research, but private pension research at large.

Establishment Specific Injury Tracking Application (ITA)

Establishments which fall into one of the 90 designated NAICS industrial categories which had more than 19 employees at any point in the previous calendar year are required to file OSHA Form 300A. This provides a repeated cross-section of establishment level health and safety outcomes. Complete years of data are publicly available from 2016 to 2020. Injuries are broken down by type, including respiratory illness, hearing loss, poisoning, skin disorder, and deaths. It also includes the NAICS code, total number of hours worked by employees, number of annual average employees, and the number of days away from work, transfers, or permanent reductions in workload associated with a safety incident. The largest overlap between these datasets is 2016 to 2019, which determines the period for analysis.

Methodology

All data cleaning and analysis in this paper is completed in R. The PPP data are merged onto OSHA observations using the employer identification numbers (EIN). For the main section of this study, only observations which were successfully merged between the two datasets are kept in the sample. Model specific restrictions will be covered in their respective sections. The dependent variable, variable of interest, and common control variables are discussed below.

Dependent Variable

The dependent variable to study health and safety outcomes is the Bureau of Labor Statistic's "Standard Incidence Rate", commonly referred to as the *incidence rate*. It is calculated as

$$R_{i,t} = \frac{\sum_k I_{i,k,t}}{H_{i,t}} * 200,000$$

where

$R_{i,t} :=$ This firm's incidence rate

$I_{i,k,t} :=$ The sum this firm's non – fatal injuries² for the period

$H_{i,t} :=$ The total hours worked by all employees at this firm for the period

The 200,000 in this formula represents 100 employees working 40 hours per week for 50 weeks, thus the incidence rate is interpreted as the employee injury rate per 100 employees. Robustness checks will also estimate a secondary workplace safety rating known as DART, which includes the days away from work, employee transfers to new roles, or permanent reduction in job capabilities resulting from an injury. These results are presented in table 1A of the appendix.

Key Explanatory Variables

The variable of interest is whether a firm has an ESOP listed as one of its pension plans, which is the dummy variable *ESOP*, set equal to 1 for an ESOP and 0 for a traditionally owned firm.

² Cases of k include injury, respiratory condition, hearing loss, poisoning, skin disorder, or other illness

Other Controls

Perhaps most importantly, the presence of a union is a necessary control, as unions are known to advocate heavily for better working conditions and may affect reporting of injuries and illnesses. Second, the Bureau of Labor Statistics has published many years of data highlighting the variation in incidence rates by industry³, so establishments are grouped using two-digit NAICS categories following the 2019 Private Pension Plan Research File User Guide⁴. I use Compustat data include a control for whether an establishment belongs to a publicly traded firm. Additionally, incidence rates may vary along with various establishment sizes, as measured by its employee base. The ITA provides the annual average number of employees per establishment per year. I log transform this value to reduce skew. Note that this is the calculated by the average number of employees per pay period over the fiscal year, so seasonal workers and short-term employees contribute to this figure. The last control is for the average weekly hours worked per employee:

$$W_{i,y} = \frac{H_{i,y}}{E_{i,y} * 50}$$

where: for year y and firm i

$W_{i,y}$ = *average weekly hours Worked per Employee*

$H_{i,y}$ = *total Hours worked by all employees at this firm for the period*

$E_{i,y}$ = *annual average number of Employees at this firm*

³ [BLS web page](https://www.bls.gov/web/osh/summ1_00.htm) for incidence rates by industries for 2020 (https://www.bls.gov/web/osh/summ1_00.htm)

⁴ [DOL web page](https://www.dol.gov/agencies/ebsa/researchers/data/private-pension-plan-data) for the Private Pension Plan Research File (<https://www.dol.gov/agencies/ebsa/researchers/data/private-pension-plan-data>)

Working more hours per week likely exposes employees to more risks, hence the inclusion.

Sample Restrictions

The following restrictions are applied after the datasets are merged and the above variables are generated. Observations are dropped from the sample if: they lack Employer Identification Number (429,426 observations dropped) or industry (20 dropped), weekly hours per employee is not between 20 and 100 hours⁵ (62,119 dropped), annual average number of employees is less than 8 (30,461 dropped), or incidence rate could not be calculated (630 dropped)

Although both datasets are cleaned and standardized by their respective agencies, the ITA dataset is difficult to work with due to reporting inconsistencies⁶. In particular, the employer identification number (EIN) was not a required reporting item until 2019, meaning that the EINs for the first three years of the dataset needed to be recovered. To do so, the ITA was also downloaded from 2020 and 2021, where the EIN is required. Establishments were then matched across all the ITA datasets, and the EINs from future years were used in places where they were not originally available. While this nearly doubled the matching rate between the ITA and PPP data points, note that every year, a small portion of business establishments are purchased by other firms, so the recovered EINs may identify that an establishment is part of a different firm prior to the year it was acquired. Given how many datapoints are in this data set and how few of these acquisitions involve ESOPs, any effect this may have should be minimal. Further, when an establishment with an ESOP is purchased by a traditional firm, the ESOP is overwhelmingly grandfathered in, which is reflected in the pension data.

⁵ Additional robustness tests will adjust this interval to explore whether this is a materially useful range.

⁶ Since the ITA's release in 2016, the BLS has published [several reports](#) documenting their exhaustive efforts to match the ITA data to their Survey of Occupational Injuries and Illness.

Restricting the weekly hours per employee and the number of annual average employees minimizes the extreme values in the sample data using reasonable restriction parameters. Both weekly hours per employee and the incidence rate are calculated using the total hours worked by all employees at an establishment. If the hours per employee value was outside of this range, the incidence rate often approached or surpassed 100 annual injuries per 100 full time workers, indicating a reporting error. Additionally, for establishments with otherwise accurate ITA entries and less than eight annual average employees, a difference of one incident across observation years would generate unrealistic variation in the incidence rate. This additional variance is a known feature of the incidence rate calculation, and the BLS has released guidance⁷ explicitly stating that the standard incidence rate calculation is more meaningful at establishments with larger employee bases.

Descriptive statistics for both the establishment and firm level restricted samples are presented in Table 1. A histogram showing the distribution of establishments across industries is presented in figure 1A. Because of idiosyncrasies regarding how companies can file their Form 5500, five publicly traded firms represent an outsized portion of the ESOPs in this sample: Target, Lowes, Publix, Wal-Mart, and Costco. These firms represent 16,531 (40%) of the ESOP establishments. That is 59.1% of the publicly traded ESOPs, and 92.1% of the ESOPs in the retail trade industry. After confirming these companies do indeed have ESOPs by checking their filings with the Securities and Exchange Commission through their EDGAR database, I have opted to leave them in the sample. Note, however, that the

[[Table 1 near here]]

⁷ <https://www.osha.gov/laws-regs/standardinterpretations/2016-08-23>. The guidance recommends that small establishments use three years of data and working hours to calculate a more realistic incidence rate. Due to working with a four-year observation window, there was no way to implement this guidance without artificially reducing the variation.

percentage of outstanding shares owned by ESOP members at publicly traded companies is normally between 5% to 15% (Chang and Mayers 1992). Accordingly, I anticipate that the ESOP's effect at these firms will be muted compared to a closely held firm with a broad-based ESOP plan. I add an interaction effect between ESOP and publicly-owned companies to decompose these effects.

Empirical Framework

The two major statistical methods I employ include a set of panel regressions and a multi-period differencing approach. The panel analysis methods are used at the firm and establishment levels to account for some of the challenges associated with matching firm level data to establishment level data, whereas the multi-period differencing is only estimated at the establishment level. The base specifications for the panel analyses are as follows:

Base Panel Specification

$$Incidence\ rate_i = \beta_0 + \beta_1 ESOP_i + \beta_2 Union_i + \beta_3 Public_i + \beta_4 (ESOP \times Public)_i + \\ Z_i' \gamma + year\ FE + industry\ FE + \varepsilon_i$$

The firm level data aggregates ITA data across all observations which share an EIN and year. Firms which share an EIN are assumed to have access to the same set of pension plans⁸. The sample restriction criteria outlined above are applied post-aggregation. The industry at the firm level is not immediately obvious because of the different business units across a firm's establishments – from corporate offices down to the manufacturing plants, retail stores, or distribution centers. I selected each firm's industry by calculating the industry with the largest

⁸ Sometimes firms which were using ESOPs get absorbed into larger companies, but this acquisition normally occurs by a publicly traded firm, which is an included control. Further, this is a small proportion of ESOP firms.

combined weight between the number of employees at each type of establishment and the number of establishments in that industry across all data years.

Each specification is estimated three times under different distributional assumptions: OLS, quantile, and censored Tobit regressions. The quantile regression is estimated on the median incidence rate and is less sensitive to the outliers present in the data. The tobit regression relaxes the strict normality assumptions of standard OLS estimation to account for the large number of firms which have zero incidents each year.

The models for each level of analysis utilize the set of valid observations which have pension data available⁹. The variance of the establishment-level model is larger than the firm-level model due to the smaller employee bases.

Differences-in-Differences Approach

An ESOP's impact on health and safety outcomes may exhibit hysteresis, necessitating a multi-period differencing approach. After adoption, the integration of participatory management practices and thus, employee-influenced decisions on health and safety, may be a several-year process. The three years between 2016 and 2019 may be sufficient time for workplace norms to reflect this participatory structure. However, estimating this model presents several challenges. The ITA data does not imply that an establishment submits an entry for every subsequent year after their first filing, generating missing years on random observations. Moreover, due to the low matching rate between the ITA and PPP data, using a balanced panel would result in low statistical power, thus suggesting the use of an unbalanced panel. Additionally, firms may adopt an ESOP in different years and may have different lengths of exposure over the course of the

⁹ Comparing firms with and without pensions is included in the robustness tests later in the paper.

observation period, creating staggered, heterogeneous treatments. The standard Two-Way/Fixed-Effects model and canonical 2x2 difference-in-difference estimation approaches have been shown to produce severe bias under such circumstances (Borusyak, Jaravel, and Spiess 2021; de Chaisemartin and D'Haultfœuille 2020; Goodman-Bacon 2021).

Accordingly, I employ the dynamically aggregated average group-time treatment estimator proposed by Callaway and Sant'Anna (2021) and implemented in R's `{did}` package. This method uses not-yet-treated observations to create a control group, allowing for an analogous multi-period parallel trends assumption. For this study, firms which adopted an ESOP in 2017, 2018, or 2019 are considered separate treatment groups, whose measured effects are then aggregated to recover power and allow for causal inference with respect to the number of years after an ESOP has been adopted. Standard errors are clustered by establishment and industry, and the untreated portion of the sample is restricted to closely resemble the treatment group's union status, rate of public ownership, and multi-establishment distribution. Considering the treatment groups are the last three observation years and there is no reason to believe that adopting an ESOP in any given year will have a differential effect on injuries and illnesses, the treatment effects should be parallel. Hence, the covariates from the panel methods above are removed here, as covariates only need to be included if different treatment groups may experience non-parallel outcomes. The third year prior to treatment is the omitted group, providing an easily interpreted estimate for the average treatment effect on the treated groups with respect to the length of treatment exposure.

Results

This section presents the main findings of the study. I am interested in whether the presence of an ESOP has any effect on the rate of injury and illness per 100 full time workers

compared to traditionally owned and managed firms. Table 2 reports the estimated coefficients for the firm-level and establishment-level regression models. The first three columns are firm-level results, and the last three are the establishment level results. Figure 1 presents predicted incidence rates in the manufacturing industry using an interaction term. Figure 2 is the result of the differences-in-differences model. I will cover each one separately in that order.

Across the three establishment-level models in the first three columns from Table 2, the coefficients for ESOPs are all negative and many have statistically significant p-values. The estimated coefficients for the OLS model, column (1), shows that the presence of an ESOP is associated with -0.92 fewer incidents per 100 full-

[[Table 2 near here]]

time workers. Incidence rates have a strong left skew and outliers are present, so a quantile regression was run on the 50th percentile, and these results are reported in column (2) of Table 2. This model estimates a larger effect of -0.934. Furthermore, there are many establishments with zero incidents across the 2016-2019 observation period, so a Tobit regression was run to account for this censored distribution. The Tobit regression in column (3) has an improved fit to the data and boasts a -1.296 reduction in incidence rates. The estimates for unionization have approximately half the magnitudes of the ESOP coefficients and all share the same sign, but the statistical significance is inconsistent. The sign, magnitude, and statistical significance of the coefficients for publicly owned companies are inconsistent across the three models. The interaction term on ESOPs and publicly owned companies consistently yields a positive coefficient across the three models, providing empirical evidence supporting the suggestion that employee-owners have less power to control their safety environment when their company is not closely held.

The firm level results of Table 2, columns (4), (5), and (6), are mostly similar to the establishment level results in the signs for ESOP, unionized, and public firms with slightly different magnitudes. The firm-level effect of an ESOP is approximately one fifth of their respective establishment-level estimates, and a union's effect at the firm level is roughly two thirds the establishment level magnitudes. The coefficients on ESOP and union are statistically significant in the quantile model, but the estimate for ESOPs loses its statistical significance in the Tobit model while the union estimate does not. These magnitudes may be muted because an ESOP or union may not cover all establishments at a firm, resulting in less influence at the firm level. Alternatively, being listed as a public firm is associated with between 1.6 and 2.2 less injuries per 100 full time employees. Perhaps optics and investor scrutiny cause public firms to keep a tight watch on health and safety protocols, and this effort is captured when the data is aggregated to the firm level. Additionally, publicly owned firms generally employ more workers than privately owned firms, so they may be more sensitive to accumulated settlement costs and production disruptions than their privately owned counterparts.

Throughout these panel results, ESOPs show a clear association with safer working environments. More broadly, these results communicate that any work environment will generally be safer when compared to a privately owned, single-location, non-unionized establishment without an ESOP, however ESOPs have the largest impact compared to these alternatives. Naturally, one might ask if a combination of these factors has an outsized effect. Accordingly, I ran twelve OLS regressions - one for each industry subset - at the establishment level with a four-way interaction between ESOPs, unions, public companies, and the number of establishments. I present five plots of predicted values for establishments in the manufacturing

industry¹⁰ where employees work between 32-48 hours per week in Figure 1¹¹. The important thing to note is that no single interaction group dominates the top or bottom of the incidence rates across all five windows.

[[Figure 1 near here]]

[[Figure 2 near here]]

Moving on to the differences-in-differences regression results presented in figure 2, this model failed to reject the parallel trends assumption (p -value = 0.36), and finds that by the second year after adoption, the incidence rates of ESOPs fall below the trend line of traditionally owned firms. The initial year of adoption also falls below the trend, but then readjusts to have no effect in the second year, implying that there may be a transition period where participatory structures are still being implemented. To test for this, I calculate the average treatment effect on the treated (ATT) and compare it against the standard two-way fixed effect (TWFE) estimate in table 2A. Indeed, the TWFE coefficient is biased toward zero because the length of exposure to the treatment is obscured in this specification. The larger magnitude in the Callaway and Sant'Anna estimate supports the idea of hysteresis playing a role regarding how ESOP adoption eventually affects workplace safety.

Study Limitations

First and foremost, OSHA's ITA website says, "this data is not representative of all businesses and general conclusions pertaining to all U.S. business should not be drawn." After working with this data for several months, the more salient drawback is the reporting inaccuracy and inconsistency borne by establishments entering their own data. Establishments may be

¹⁰ The manufacturing industry was selected for this interaction analysis because it has the most observations across the interaction cells.

¹¹ The full, untruncated table of interaction-term coefficients is available from the author upon request

directed to file if their local authorities observe elevated injury rates, potentially biasing the ITA toward more incident-prone establishments. These concerns are lessened with the pre/post longitudinal analysis. Either way, the low matching rate and sample restrictions may have introduced some amount of sample bias.

Matching one dataset from the firm level to another at the establishment level presents unique challenges. My attempts to improve the matching covered three domains. One, a fuzzy matching algorithm to link the ITA establishment names and addresses to specific pension plans in the PPP data. Two, mirroring probabilistic matching techniques developed by the BLS to match ITA data to their SOII data. Three, a tracing algorithm to recover correct EINs by mapping out firms which changed their EIN in the PPP data. Despite these efforts, the matches may miss a number of firms.

Several attempts were made to construct a firm's "ESOP participation rate" to better understand how much influence the plan members might exert on managerial decision-making. Getting a clearer picture of how involved ESOP employees are in managerial decision-making is important for understanding the mechanisms at work.

The panel regression methods likely exhibit some amount of omitted variable bias. Less bias could be achieved by gathering more in-depth financial and human resource data. For example, having a better measure of the distribution of shares and the total percentage of shares outstanding owned across ESOP employees would allow these models to estimate the relationship between participation and workplace safety. Data that describes how well advertised an ESOP plan is or whether there are training programs for managerial decision making and involvement would accomplish a similar task. Better financial data may help indicate which

firms have recently purchased new safety equipment, provided health and safety training, or had better turnover rates.

The standard errors on the quantile regressions are not robust to heteroscedasticity. Bootstrapping would be appropriate to generate more accurate standard errors.

Conclusion

ESOP establishments have fewer injuries and illnesses than otherwise-similar non-ESOP establishments. The panel models in Table 2 have magnitudes ranging from 0.92 to 1.296 fewer incidents per 100 full time employees at the establishment level depending on the model, equivalent to a 12-45% reduction in an establishment's incidence rate. An ESOP's effect on workplace safety is consistently larger than that of a union, and complementary research referenced throughout this paper suggests the above-market compensation associated with ESOPs is also comparable to the increased pay provided by union bargaining. The interaction analysis revealed that an ESOP is not necessarily a "silver bullet" to improve workplace safety. Instead, its association with safe working conditions varies slightly by industry, number of establishments, unionization, and whether it is publicly owned. From a causal perspective, I find that the adoption of an ESOP may lead to lower incidence rates after two years of adjusting to an employee-owner based workforce. This hysteresis may be explained by a combination of the vesting period for employees to gain voting rights, their bargaining power requiring enough time to reach a critical mass, and the challenges of implementing participatory channels.

The research presented here combined with the existing literature suggests that ESOPs may provide simultaneous solutions for stagnant wages, sluggish productivity, and improved

workplace safety. Employees, firms, and public health experts should keep these results in mind as they select their workplace, seek performance improvements, and write policy.

Future research in this area could include the effects of employee ownership on other nonpecuniary benefits such as flexibility of working hours, employment stability, and psychological health. This study opens the door for exploring a more-complete set of nonpecuniary analysis in ESOP firms. The interaction analysis by industry revealed that workplace safety is not necessarily associated with ESOPs across every domain, but perhaps there are nonpecuniary benefits in specific industries which matter more depending on the profession. For example, manufacturing and construction are naturally more hazardous industries than finance. The former may care about workplace safety, while the latter may show more interest in nonpecuniary benefits such as childcare or flextime scheduling. In addition, future research would be valuable on how employee ownership affects norms of fairness and cooperation that might be an important mechanism explaining its relationship to workplace safety.

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Table 1: Descriptive Statistics

Characteristic	N	Establishment		Firm	
		ESOP	Traditional	ESOP	Traditional
		40,889 ^I	212,044 ^I	3,166 ^I	80,109 ^I
Incidence Rate		4.3 (4.2)	4.7 (6.0)	4.0 (4.3)	4.7 (5.3)
Unionized		8,071 (19.7%)	35,252 (16.6%)	403 (12.7%)	3,743 (4.7%)
Public		27,990 (68.5%)	26,714 (12.6%)	470 (14.8%)	1,947 (2.4%)
Annual Average Employees		188 (535)	150 (616)	2,462 (17,884)	401 (4,730)
Weekly Hours per Employee					
20-35 hours		24,195 (59.2%)	71,891 (33.9%)	448 (14.2%)	17,299 (21.6%)
35-50 hours		15,937 (39.0%)	129,930 (61.3%)	2,613 (82.5%)	58,983 (73.6%)
50-65 hours		708 (1.7%)	9,042 (4.3%)	97 (3.1%)	3,486 (4.4%)
65-80 hours		49 (0.1%)	1,181 (0.6%)	8 (0.3%)	341 (0.4%)
Industry					
Agriculture		220 (0.5%)	3,008 (1.4%)	12 (0.4%)	1,477 (1.8%)
Construction		1,617 (4.0%)	26,352 (12.4%)	769 (24.3%)	19,845 (24.8%)
Finance, insurance & real estate		161 (0.4%)	3,847 (1.8%)	21 (0.7%)	846 (1.1%)
Information		302 (0.7%)	864 (0.4%)	6 (0.2%)	209 (0.3%)
Manufacturing		7,345 (18.0%)	55,344 (26.1%)	1467 (46.4%)	31,940 (40.0%)
Mining		33 (0.1%)	657 (0.3%)	11 (0.3%)	342 (0.4%)
Misc. organizations		2 (0.0%)	440 (0.2%)	0 (0.0%)	158 (0.2%)
Retail trade		24,081 (58.9%)	37,497 (17.7%)	181 (5.7%)	3,311 (4.1%)
Services		1,255 (3.1%)	41,708 (19.7%)	255 (8.1%)	11,765 (14.7%)
Transportation		1,191 (2.9%)	17,779 (8.4%)	73 (2.3%)	3,474 (4.3%)
Utilities		2,338 (5.7%)	2,407 (1.1%)	87 (2.7%)	586 (0.7%)
Wholesale trade		2,344 (5.7%)	22,141 (10.4%)	282 (8.9%)	6,620 (7.8%)
N establishments					
	1			1,930 (61.0%)	62,762 (78.3%)
	2-5			636 (20.1%)	11,699 (14.6%)
	6-19			356 (11.2%)	4,237 (5.3%)
	50-249			102 (3.2%)	438 (0.5%)
	20-49			121 (3.8%)	931 (1.2%)
	250-999			10 (0.3%)	36 (0.0%)
	Greater than 1000			11 (0.3%)	6 (0.0%)

This table contains the descriptive statistics for the panel sample stratified by establishments and firms. An observation is an ESOP if an ESOP is indicated among any of that firm's pension plans. The sample is restricted to observations which relevant control variables could be calculated and whose average weekly hours per employee were between 20 and 80 hours per week. Only observations which could be matched across the two data sources were kept.

^I Mean (SD); n (%)

Pension Data comes from the IRS Form 5500 Research File. Health and safety data comes from OSHA's Incidence Tracker App. Both datasets were imported for the years 2016 thru 2019

Table 2: Relationship Between ESOP presence and Incidence Rates

Dependent variable: Annual Injuries Per 100 Full-time Employees						
	Establishments			Firms		
	<i>panel</i>	<i>quantile</i>	<i>Tobit</i>	<i>panel</i>	<i>quantile</i>	<i>Tobit</i>
	<i>linear</i>	<i>regression</i>		<i>linear</i>	<i>regression</i>	
	(1)	(2)	(3)	(4)	(5)	(6)
ESOP	-0.920* (0.257)	-0.934*** (0.030)	-1.296*** (0.359)	-0.179 (0.189)	-0.266*** (0.066)	-0.223 (0.186)
Unionized	-0.506 (0.377)	-0.470*** (0.019)	-0.594 (0.420)	-0.320 (0.152)	-0.303*** (0.045)	-0.509*** (0.138)
Public	-0.099 (0.508)	-0.402*** (0.026)	0.242 (0.660)	-1.685** (0.249)	-1.655*** (0.052)	-2.247*** (0.205)
ESOP x Public	0.728 (0.871)	0.769*** (0.042)	0.906 (1.042)	0.034 (0.273)	0.112 (0.092)	0.025 (0.341)
Constant	7.537 (2.915)	2.325*** (0.146)	2.802 (0.259)	13.208** (1.214)	9.091*** (0.317)	10.638*** (1.410)
Industry dummies (2-digit NAICS)	yes	yes	yes	yes	yes	yes
Year dummies (2016-2019)						
Clustered SE	yes	no	yes	yes	no	yes
Adjusted R ²	0.030			0.034		
Observations	252,933	252,933	252,933	83,336	83,336	83,336

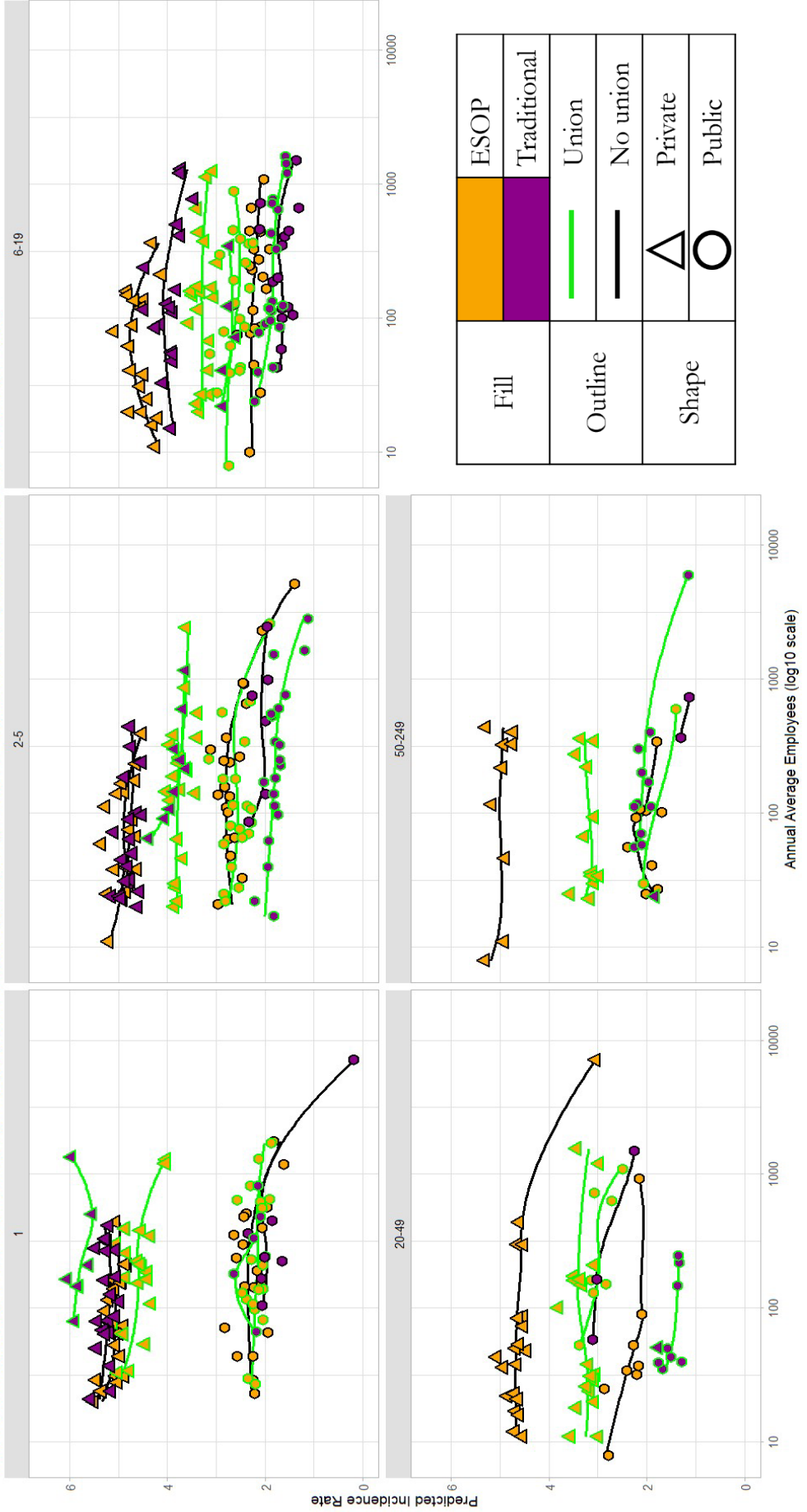
Note: Standard errors in parentheses. This table contains the coefficients describing the relationship between the presence of an ESOP and the incidence rate for six multivariate regression models, 1-3 at the establishment level, and 4-6 at the firm level. The quantile regressions were estimated on the median. The base group for ESOP presence is non-ESOP firms. Observations were dropped if the establishment had less than 8 annual average employees or the weekly hours per employee was not between 20 and 80. Fixed effects consist of industry and year estimates. Standard errors were clustered at the industry and year level where indicated. There are 12 industry dummies.

Data comes from OSHA's Injury Tracking App and is combined with IRS Private Pension Plan data for the years 2016 through 2019.

* (p<0.1); ** (p<0.05); *** (p<0.01)

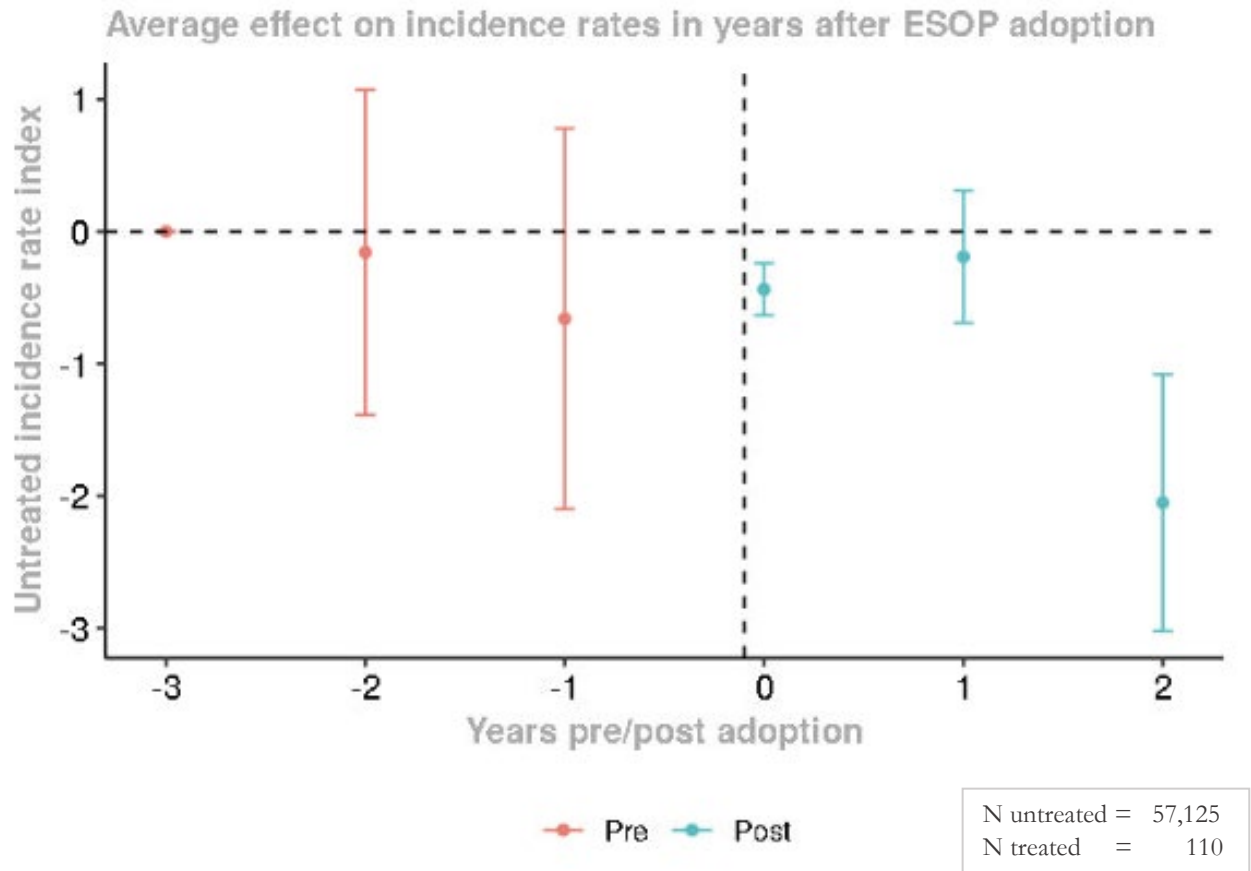
Figure 1

Predicted Incidence Rates in the Manufacturing Industry (with interactions, grouped by number of establishments, restricted between 32-48 hours per week)



Note: These plots contain predicted values from an OLS regression on establishments in the manufacturing industry with a four-way interaction on the number of establishments, public status, unionization, and ESOP presence. It is stratified by the number of establishments, indicated in the gray bars at the top of each plot. The number of annual average employees is on the x-axis with a log10 scale. To keep these plots readable, a maximum of 20 establishments were randomly sampled from each interaction group.

Figure 2



Note: Staggered treatments were used, meaning that an ESOP could have been adopted in 2017, 2018, or 2019, and was included so long as at least one pre-treatment year was available and the establishment did not drop the ESOP in subsequent years. Standard errors were clustered by establishment and industry. This plot shows the dynamic effect of exposure to an ESOP. The third year prior to treatment is the omitted group. Confidence bands are at the 95% level. The vertical dashed line separates pre and post periods.

Appendix

Section 1

This plot is summarized from the establishment level data used in this study.

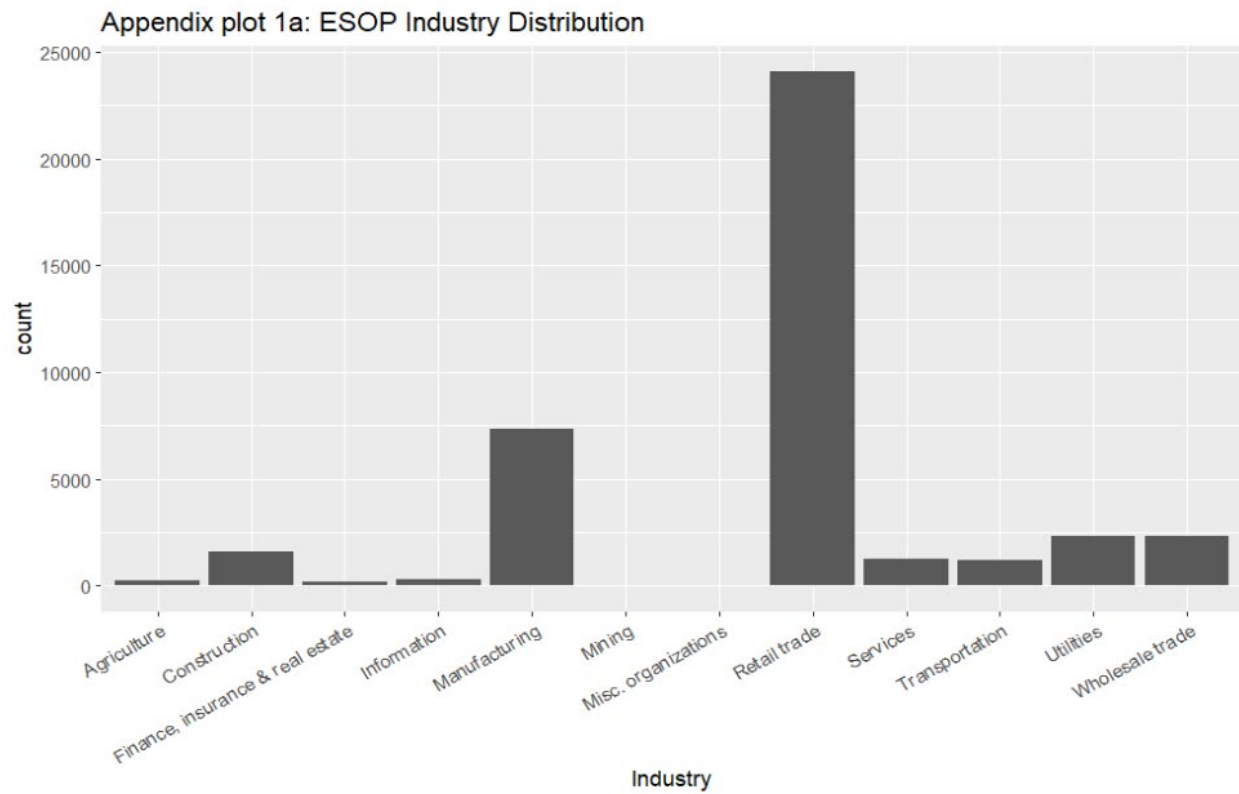


Table 1A: Establishment Level Relationship Between ESOP presence and DART rates

	<i>DART</i>		
	<i>panel</i>	<i>quantile</i>	<i>Tobit</i>
	<i>linear</i> (1)	<i>regression</i> (2)	(3)
ESOP	-0.652 (0.219)	-0.395*** (0.018)	-1.114*** (0.374)
Unionized	-0.217 (0.293)	-0.204*** (0.012)	-0.328 (0.369)
Public	0.142 (0.388)	-0.117*** (0.016)	0.108 (0.576)
ESOP x Public	0.576 (0.792)	0.435*** (0.025)	0.836 (1.138)
Constant	3.094 (2.205)	1.457*** (0.065)	3.211 (2.802)
Fixed Effects	yes	yes	yes
Clustered SE	yes	no	yes
Quantile		0.5	
Adjusted R ²	0.026		
Observations	252,933	252,933	252,933

Table 2A: Comparing ATT estimates

Estimand	Two-way Fixed Effects	Callaway and Sant'Anna
Coefficient	-0.435***	-0.522***
Standard error	(0.126)	(0.167)

Note: both specifications clustered standard errors by establishment and industry